Edu Tutor AI: Personalized Learning

Generative AI with IBM

Project Description:

EduTutor AI uses the Granite model from Hugging Face to create simple, personalized learning tools like concept explainers, quizzes generator and add more functionalities that you like. This project is deployed in Google Colab using Granite for low setup effort and reliable performance.

Pre-requisites:

1. Gradio Framework Knowledge: Gradio Documentation
2. IBM Granite Models (Hugging Face): IBM Granite models
3. Python Programming Proficiency: Python Documentation
4. Version Control with Git: Git Documentation
5. Google Collab's T4 GPU Knowledge: Google collab

Project Workflow:

Activity-1: Exploring Naan Mudhalavan Smart Interz Portal.

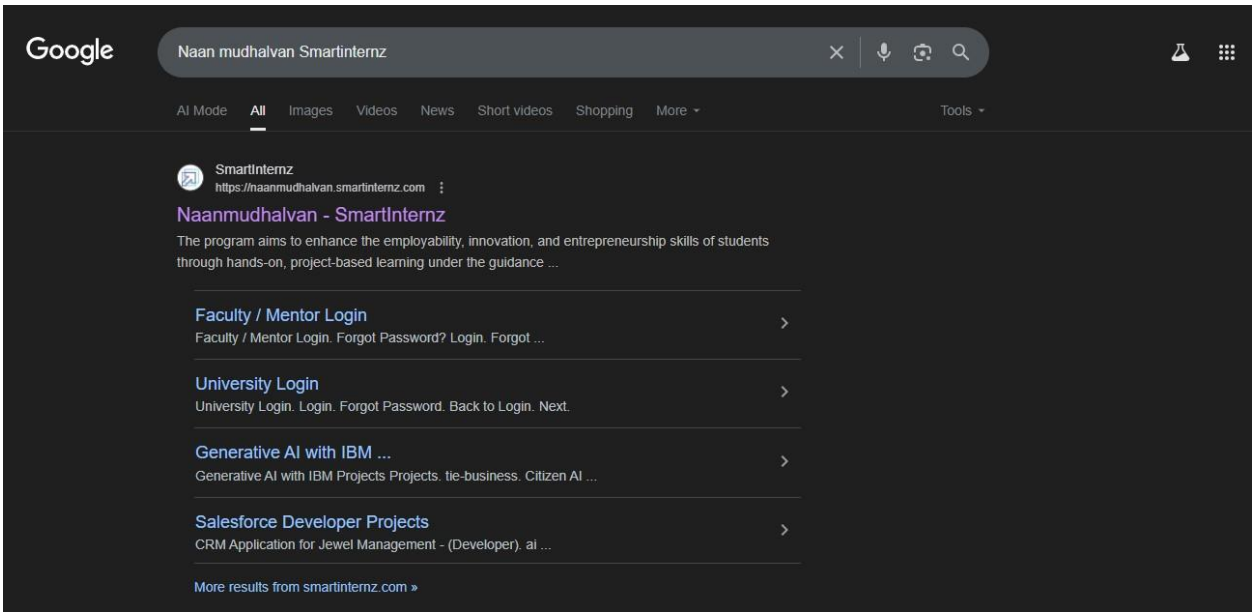
Activity-2: Choosing a IBM Granite Model From Hugging Face.

Activity-3: Running Application In Google Colab.

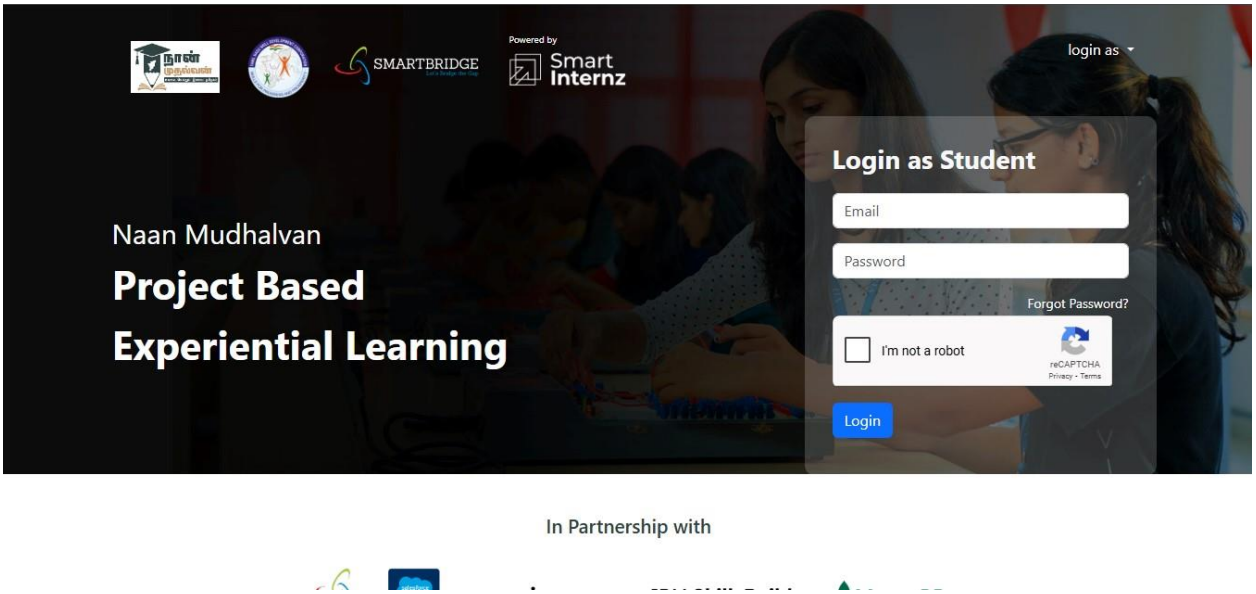
Activity-4: Upload your Project in Github.

Activity-1: Exploring Naan Mudhalavan Smart Interz Portal.

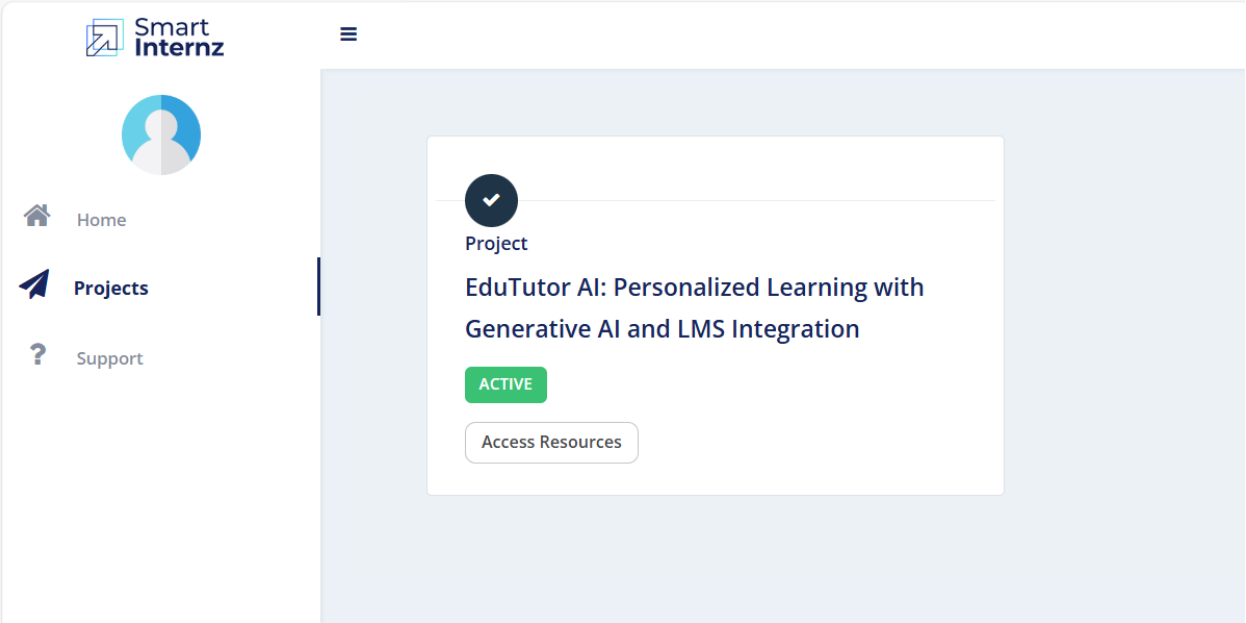
- Search for "Naan Mudhalavan Smart Interz" Portal in any Browser.



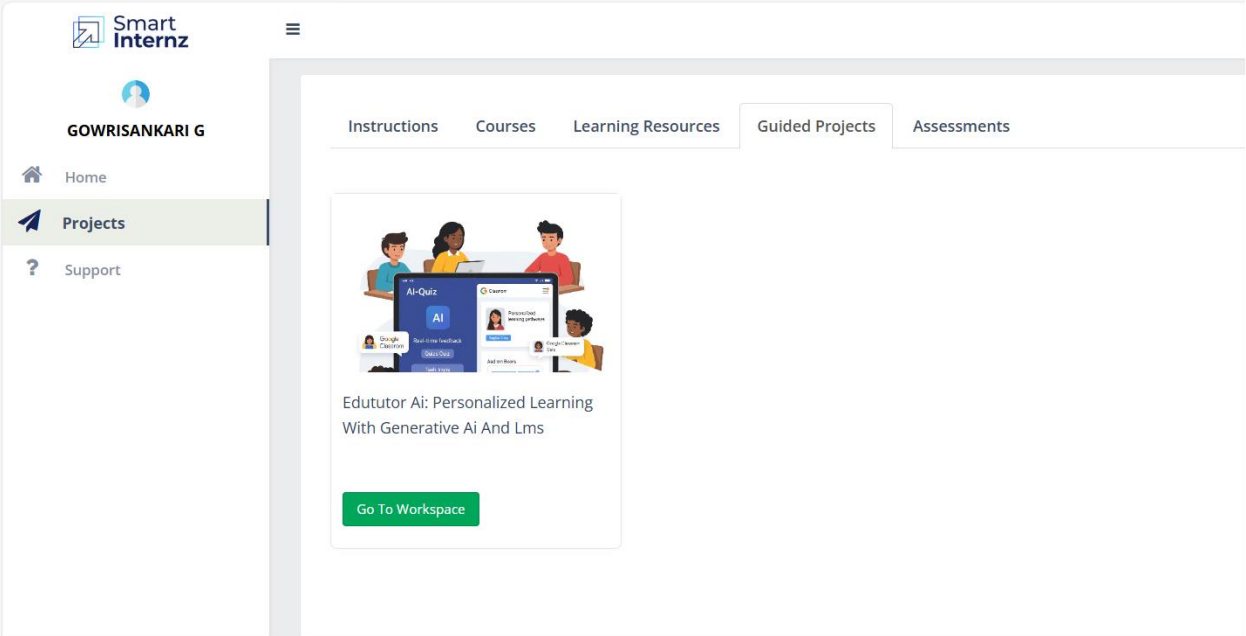
- Then Click on the first link. (Naanmudhalvan Smartinternz) Then login with your details.



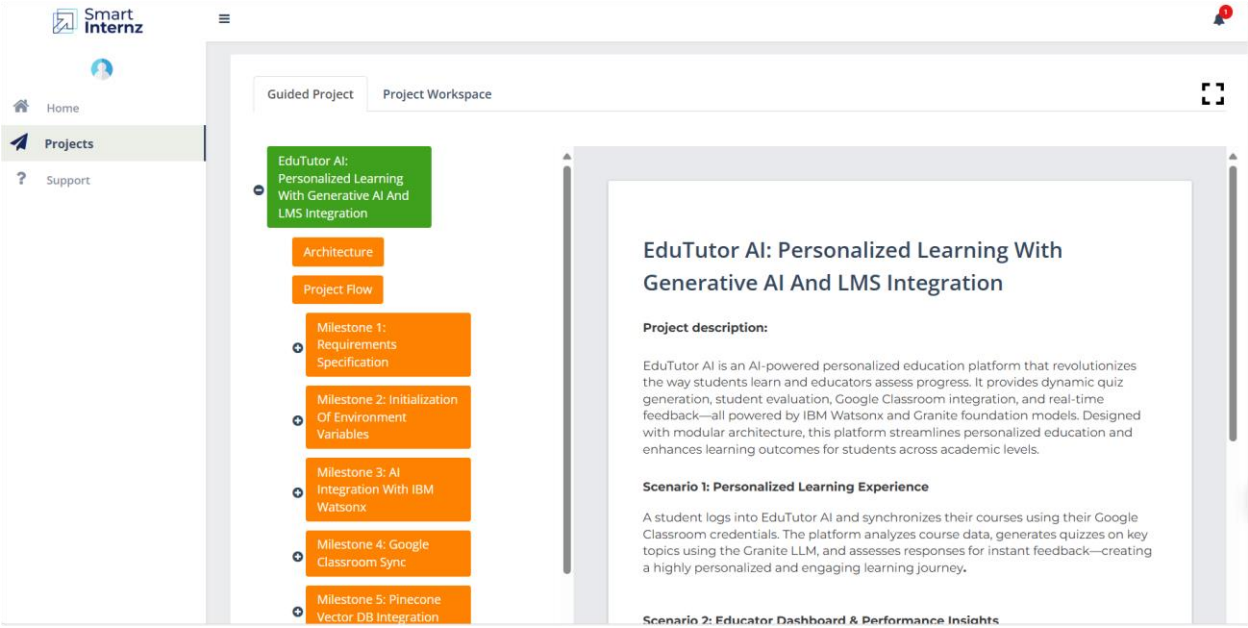
- Then you will be redirected to your account then click on “Projects” Section. There you can see which project you have enrolled in here it is “EduTutor AI”.



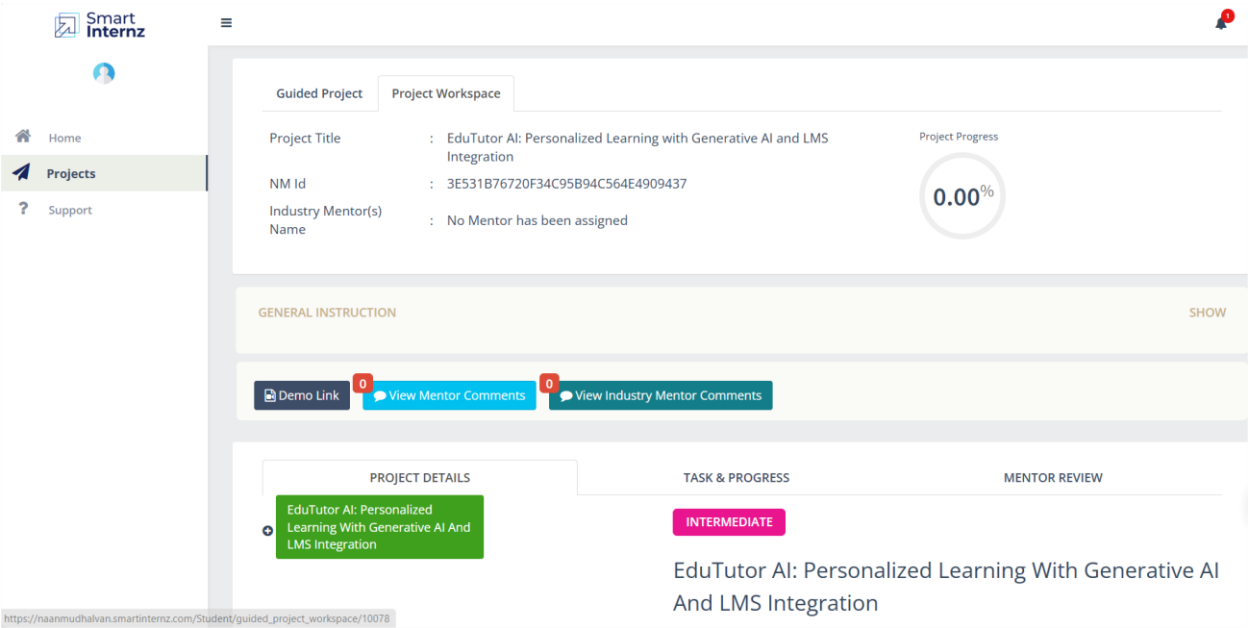
- Then click on “Access Resources” and go to the “Guided Project” Section.



- Click on the “Go to workspace” section. Then you can find the detailed explanation of Generative AI Project using IBM Watsonx API key.



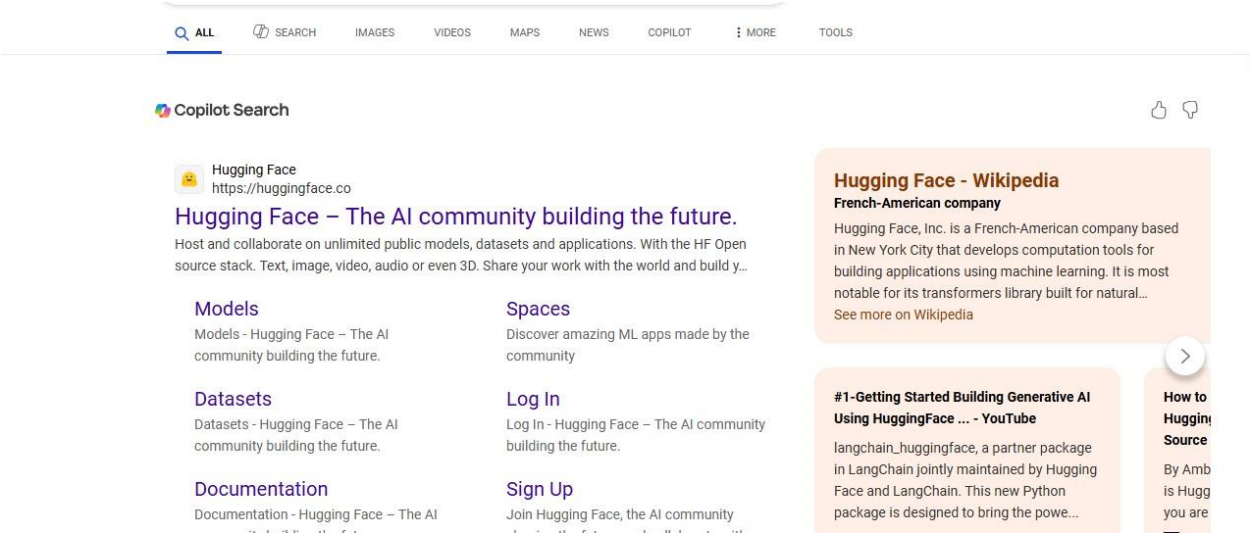
- Click on “Project Workspace”, there you can find your project progress and Place to upload “Demo link”.



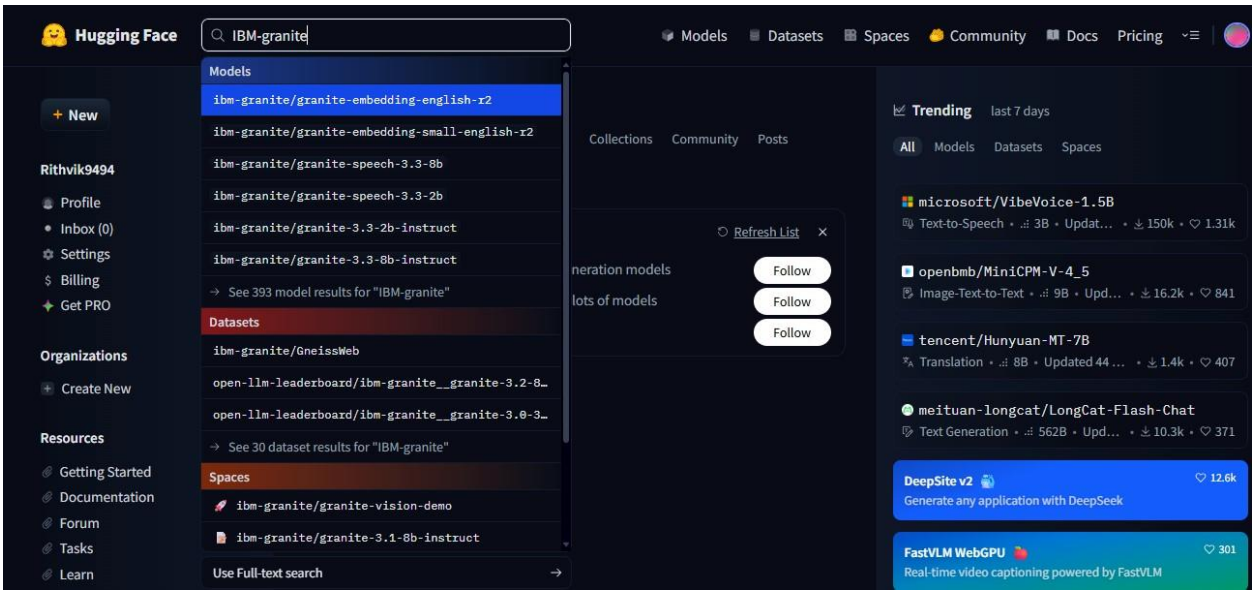
- Now we have gone through portal understanding, now lets find a IBM granite model from hugging face to integrate in our project.

Activity-2: Choose a IBM Granite model From Hugging Face.

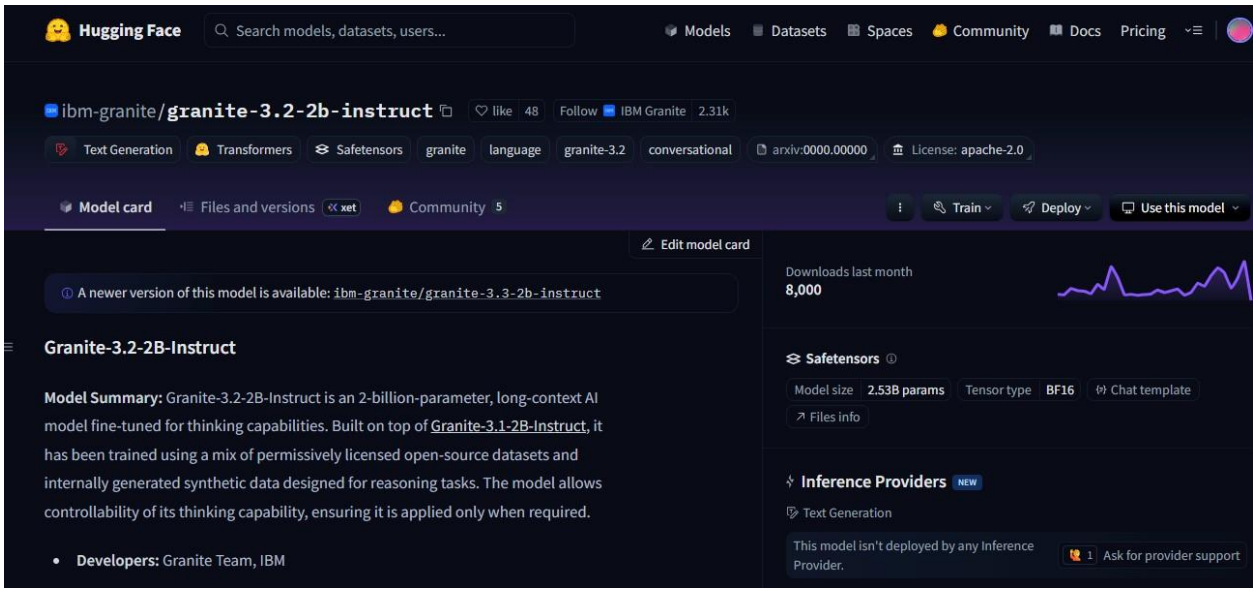
- Search for “Hugging face” in any browser.



- Then click on the first link (Hugging Face), then click on signup and create your own account in Hugging Face. Then search for “IBM-Granite models” and choose any model.



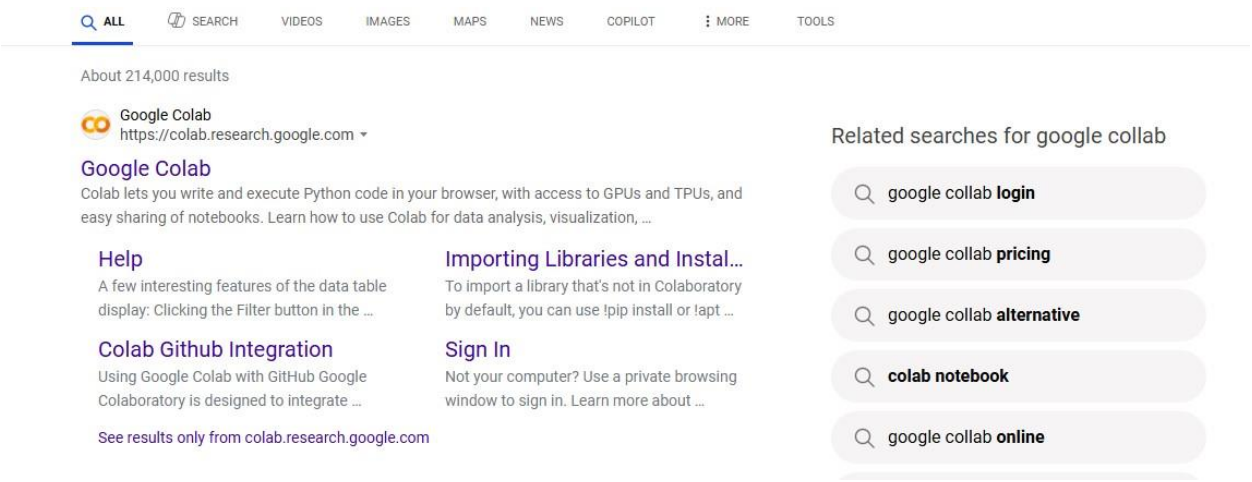
- Here for this project we are using “granite-3.2-2b-instruct” which is compatible fast and light weight.



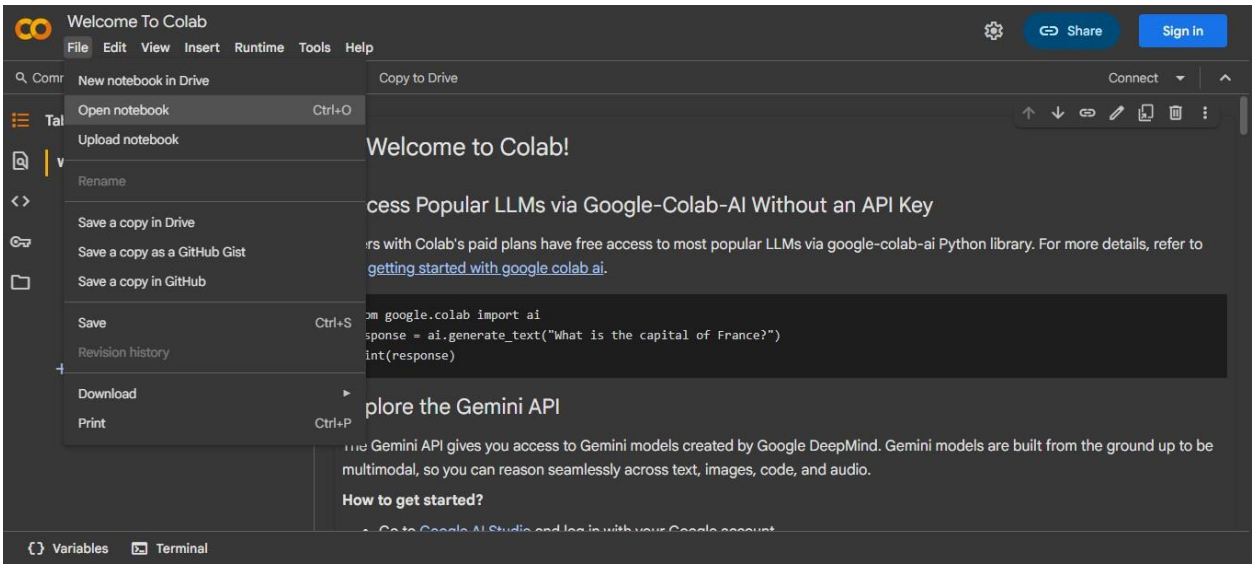
- Now we will start building our project in Google collab.

Activity-3: Running Application in Google Collab.

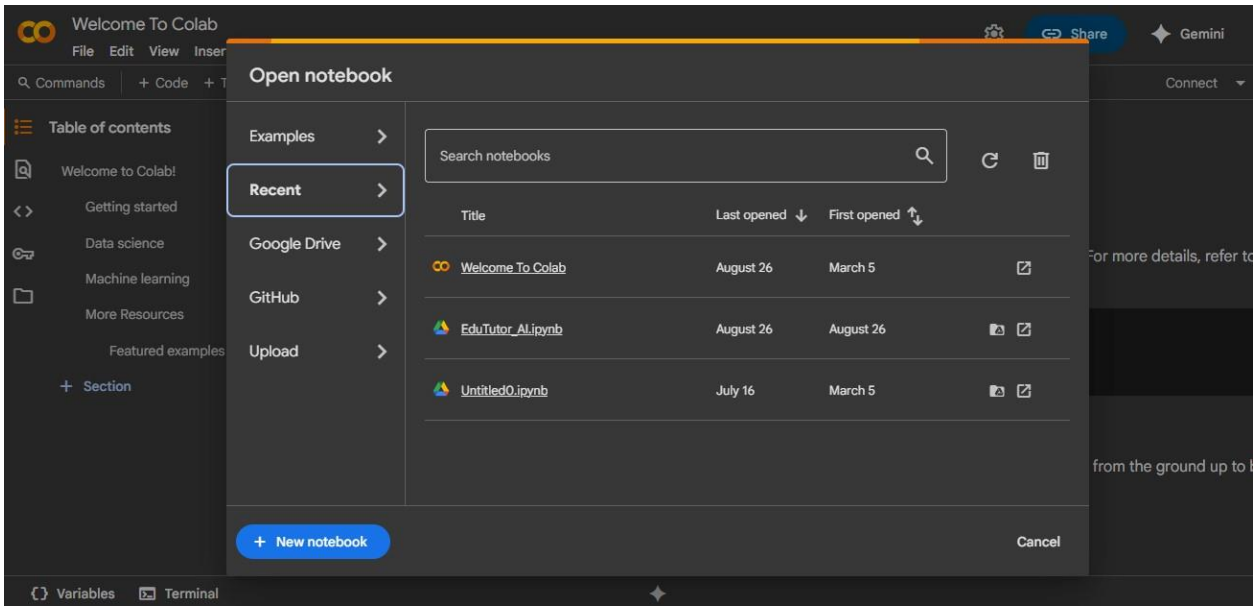
- Search for “Google collab” in any browser.



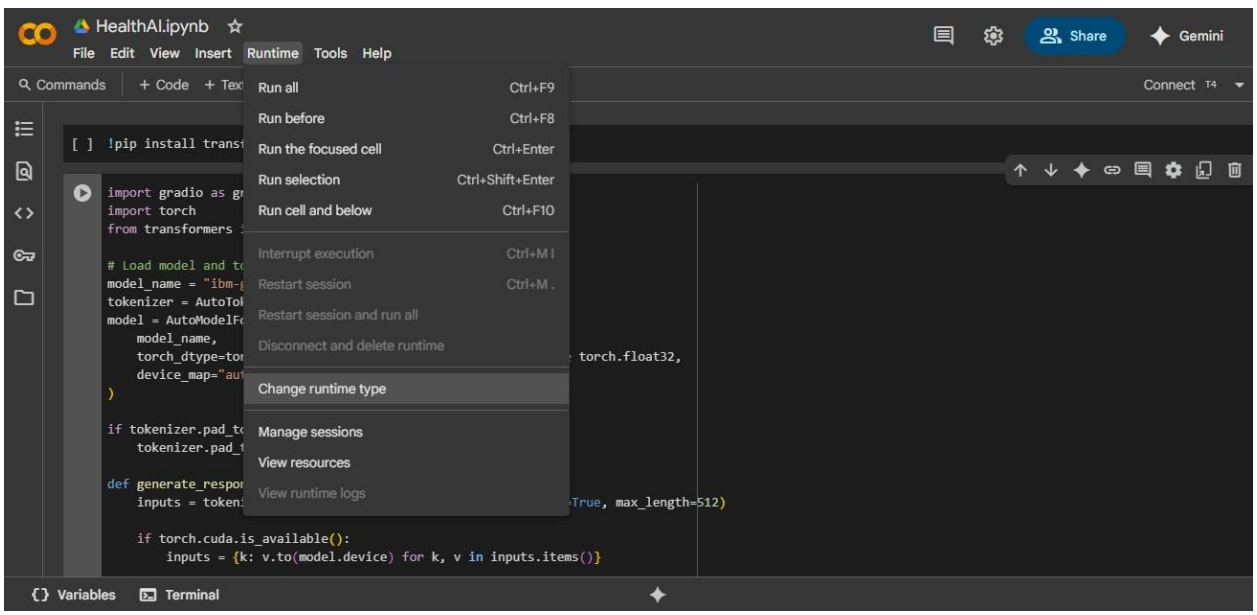
- Click on the first link (Google Colab), then click on “Files” and then “Open Notebook”.



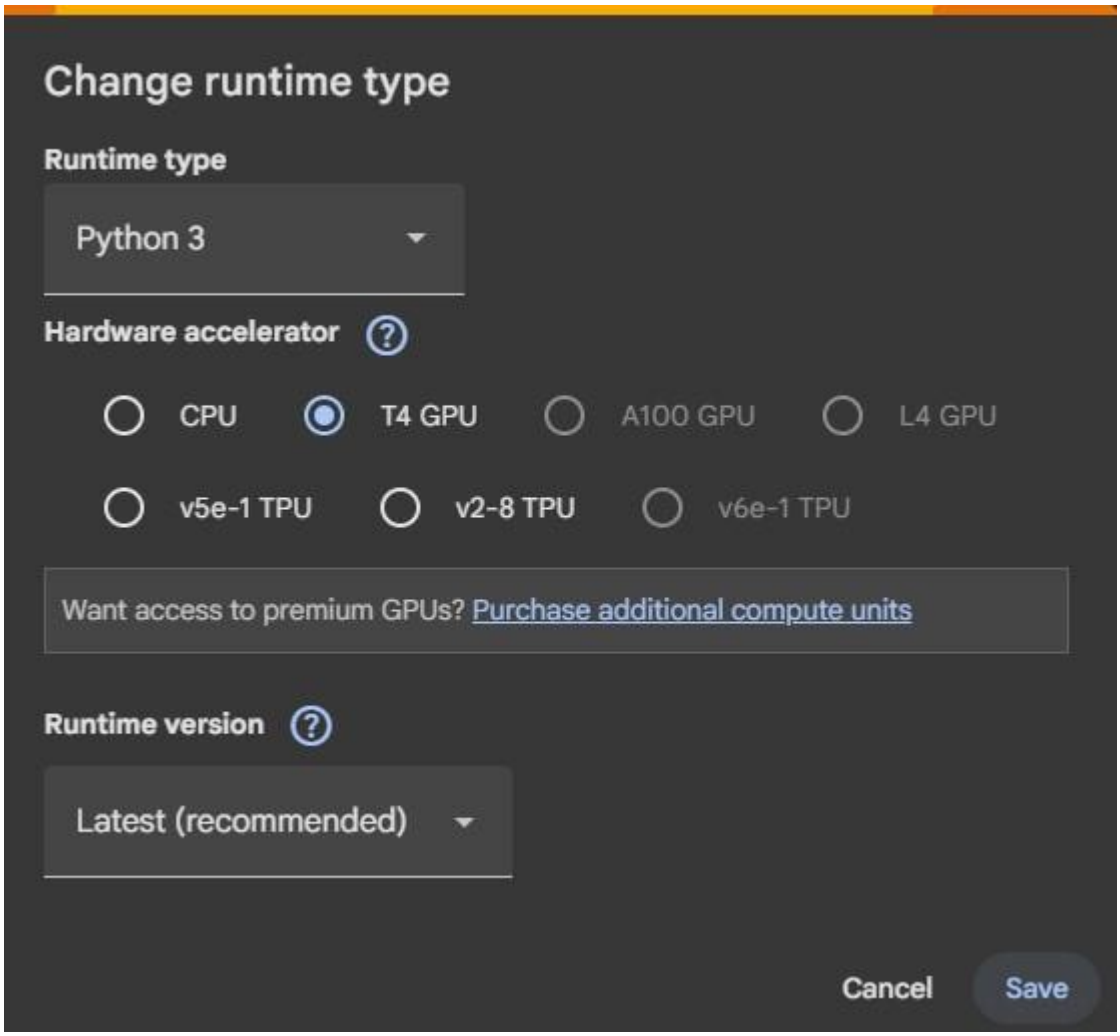
- Click on “New Notebook”



- Change the title of the notebook “Untitled” to “Health AI”. Then click on “Runtime”, then go to “Change Runtime Type”.

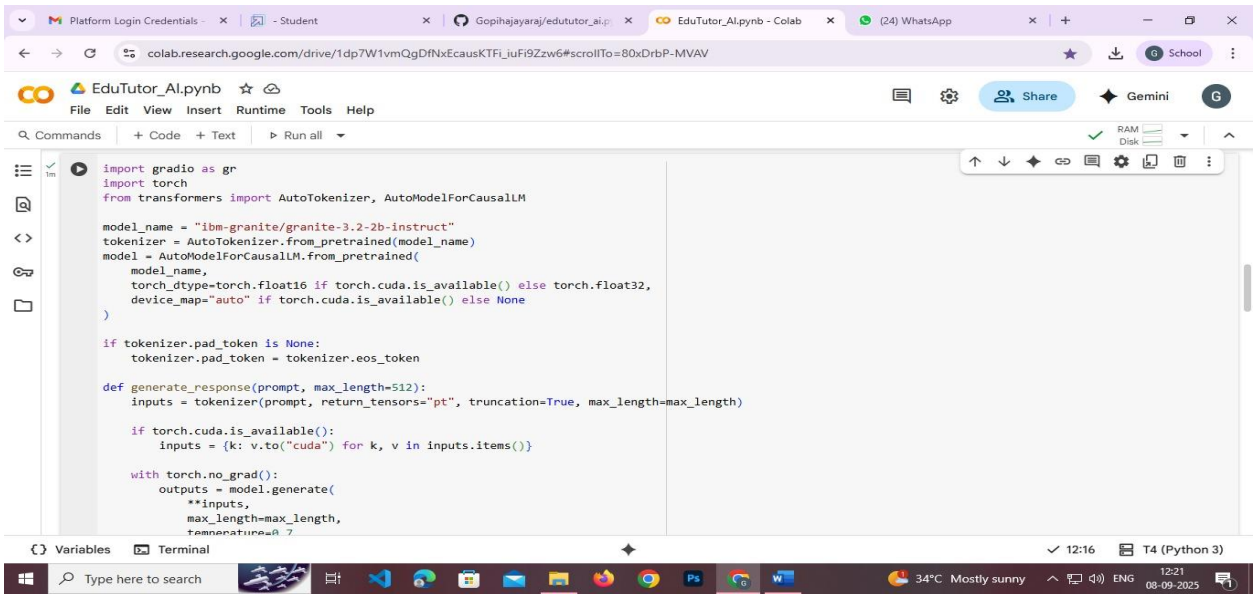


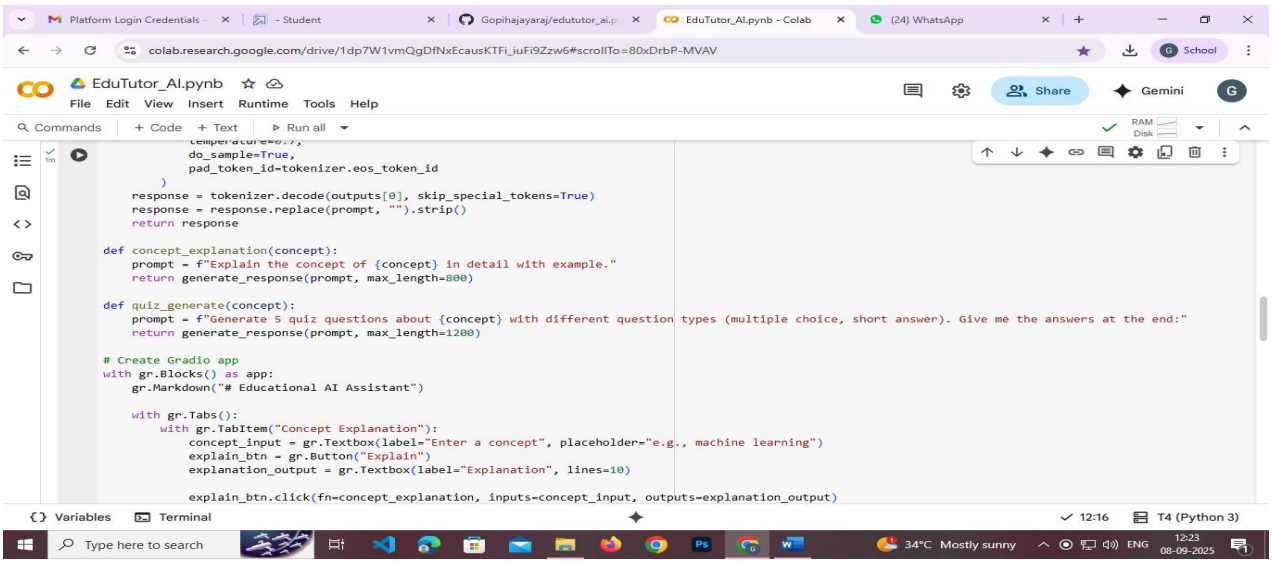
- Choose “T4 GPU” and click on “Save”

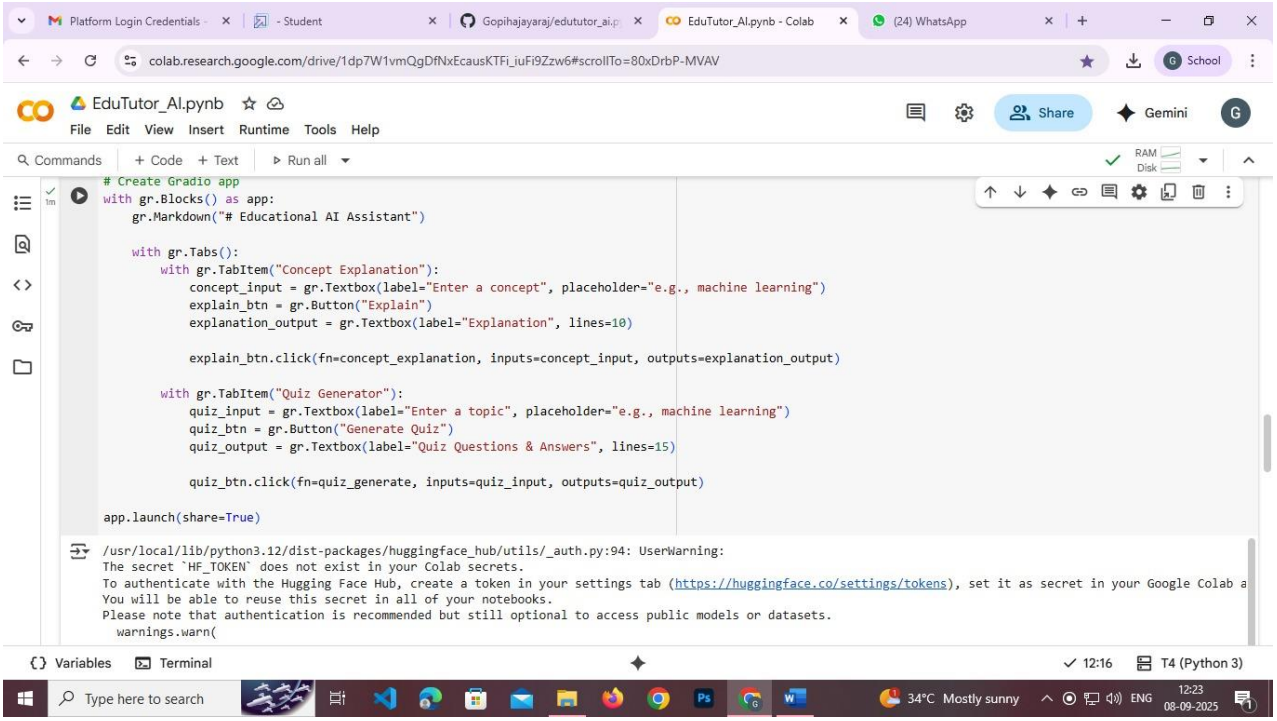


- Then run this command in the first cell "`!pip install transformers torch gradio -q`". To install the required libraries to run our application.

- Then run the rest of the code in the next cell.


EduTutor_AI.pynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
import gradio as gr
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM
model_name = "ibm-granite/granite-3.2-2b-instruct"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(
model_name,
torch_dtype=torch.float16 if torch.cuda.is_available() else torch.float32,
device_map="auto" if torch.cuda.is_available() else None
)
if tokenizer.pad_token is None:
tokenizer.pad_token = tokenizer.eos_token
def generate_response(prompt, max_length=512):
inputs = tokenizer(prompt, return_tensors="pt", truncation=True, max_length=max_length)
if torch.cuda.is_available():
inputs = {k: v.to("cuda") for k, v in inputs.items()}
with torch.no_grad():
outputs = model.generate(
**inputs,
max_length=max_length,
Variables Terminal
12:16 T4 (Python 3)

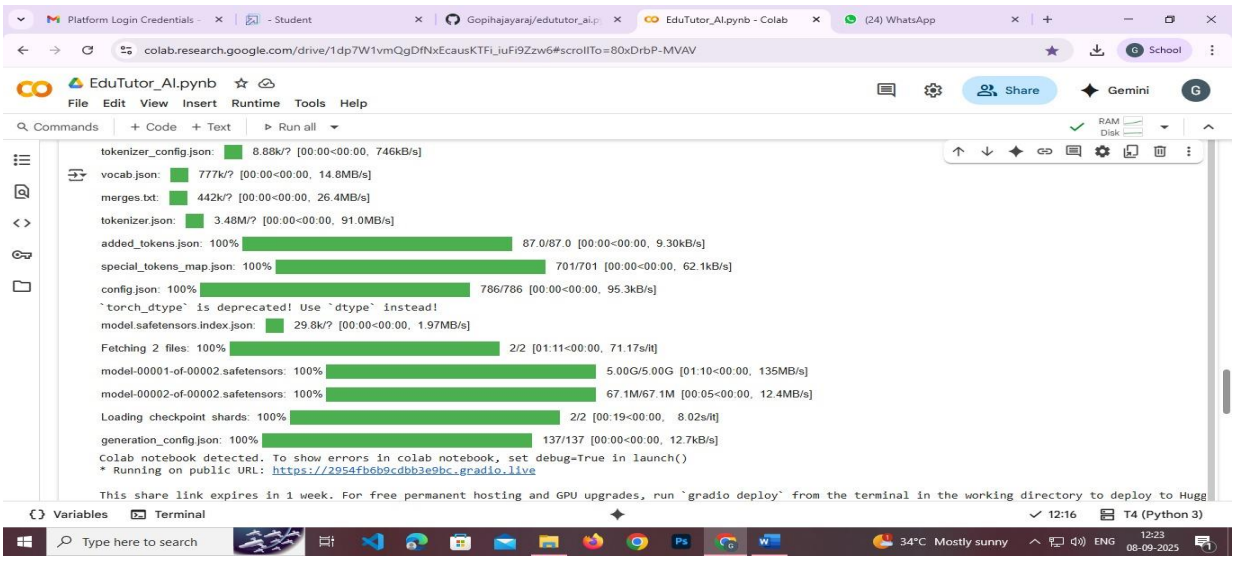

EduTutor_AI.pynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
do_sample=True,
pad_token_id=tokenizer.eos_token_id
response = tokenizer.decode(outputs[0], skip_special_tokens=True)
response = response.replace(prompt, "").strip()
return response
def concept_explanation(concept):
prompt = f"Explain the concept of {concept} in detail with example."
return generate_response(prompt, max_length=800)
def quiz_generate(concept):
prompt = f"Generate 5 quiz questions about {concept} with different question types (multiple choice, short answer). Give me the answers at the end:"
return generate_response(prompt, max_length=1200)
Create Gradio app
with gr.Blocks() as app:
gr.Markdown("# Educational AI Assistant")
with gr.Tabs():
with gr.TabItem("Concept Explanation"):
concept_input = gr.Textbox(label="Enter a concept", placeholder="e.g., machine learning")
explain_btn = gr.Button("Explain")
explanation_output = gr.Textbox(label="Explanation", lines=10)
explain_btn.click(fn=concept_explanation, inputs=concept_input, outputs=explanation_output)
Variables Terminal
12:16 T4 (Python 3)



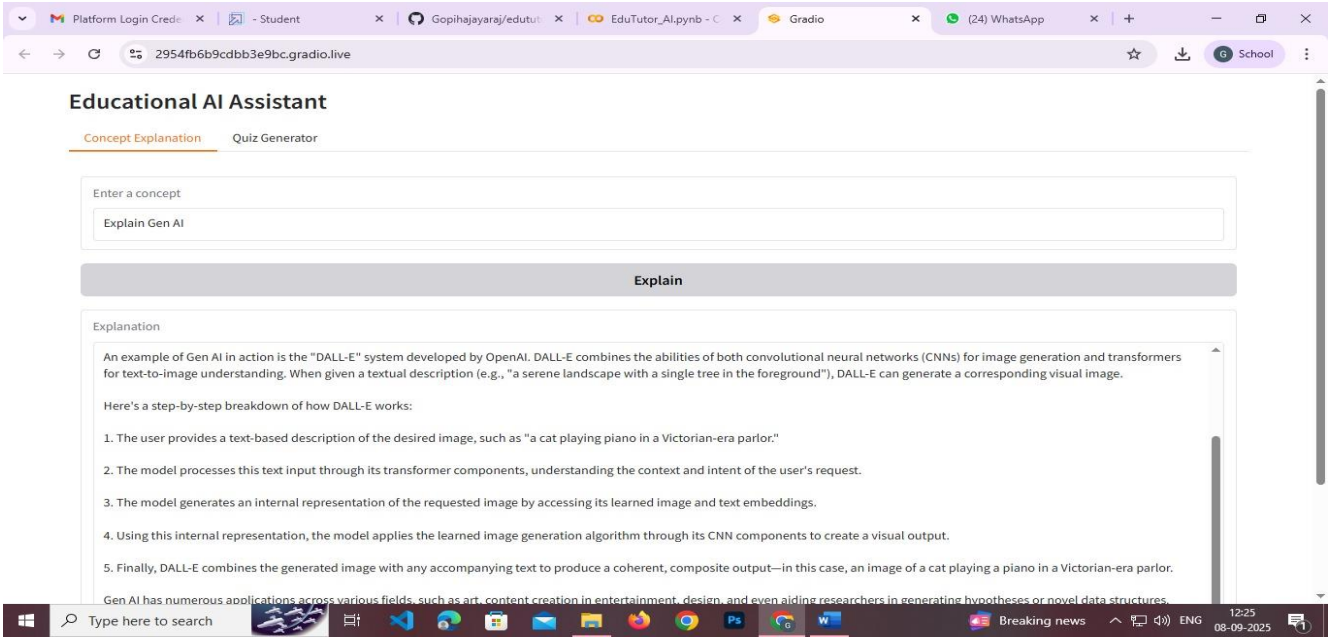
- You can find the code here in this link: [Edu Tutor AI Code]

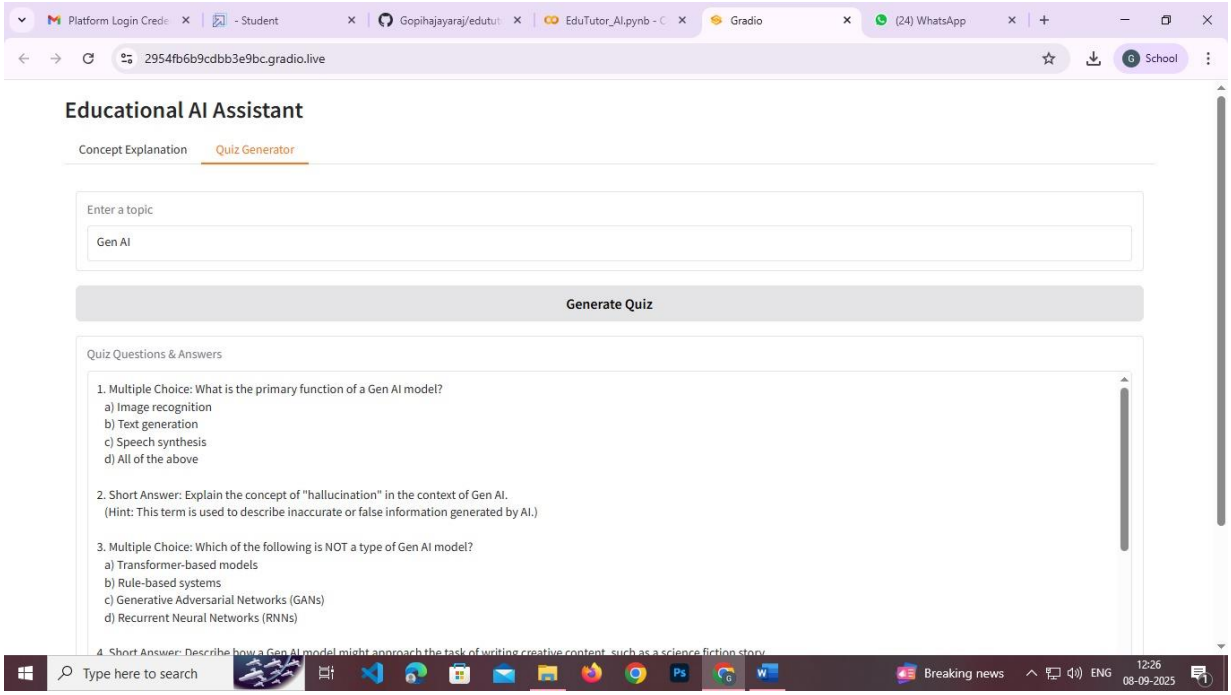
OUTPUT:

- Now you can see our model is being Downloaded and the application is running.



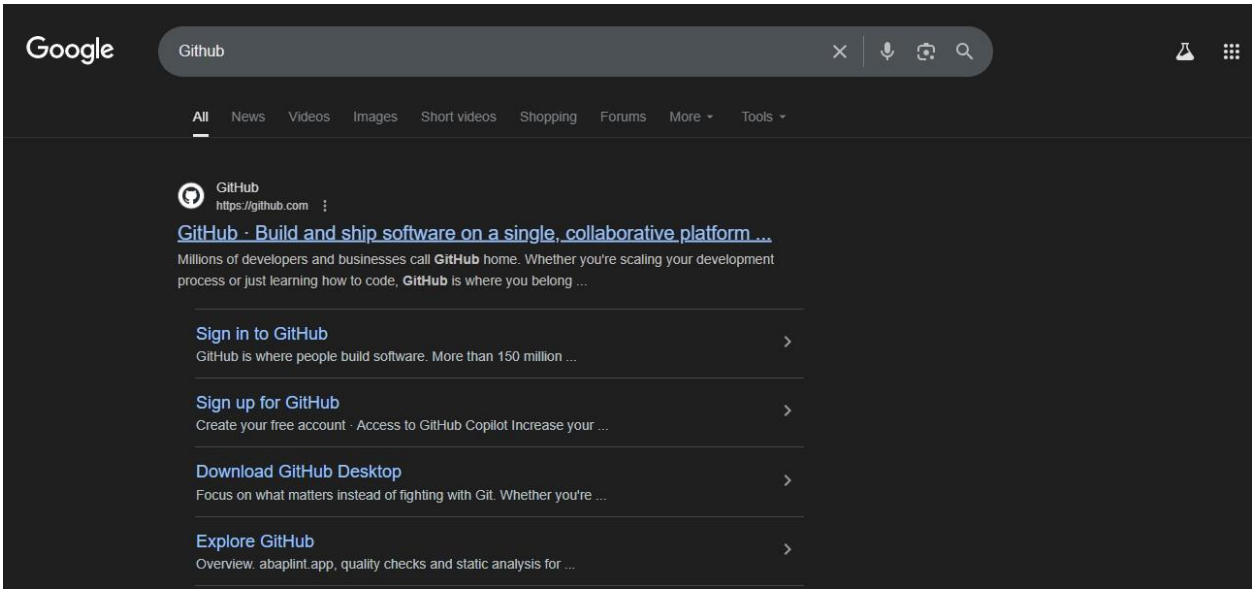
- Click on the URl to open the Gradio Application click on the link.
- You can View the Application is running in the other tab.



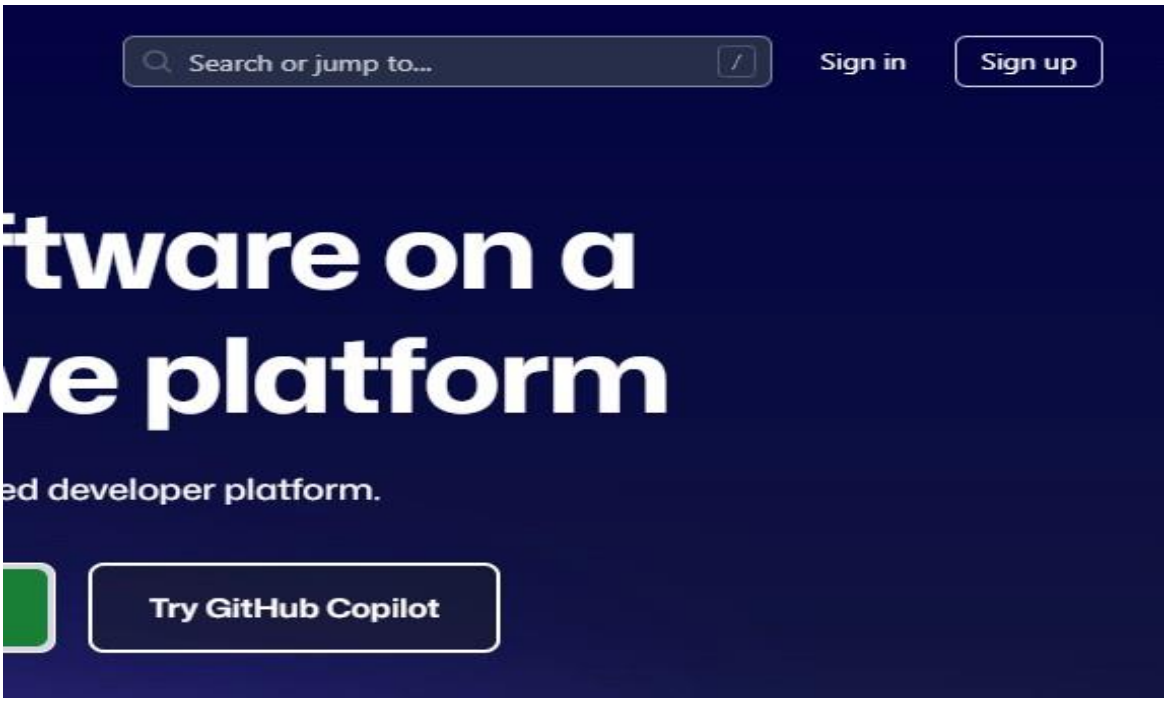


Activity-4: Upload Your Project in GitHub.

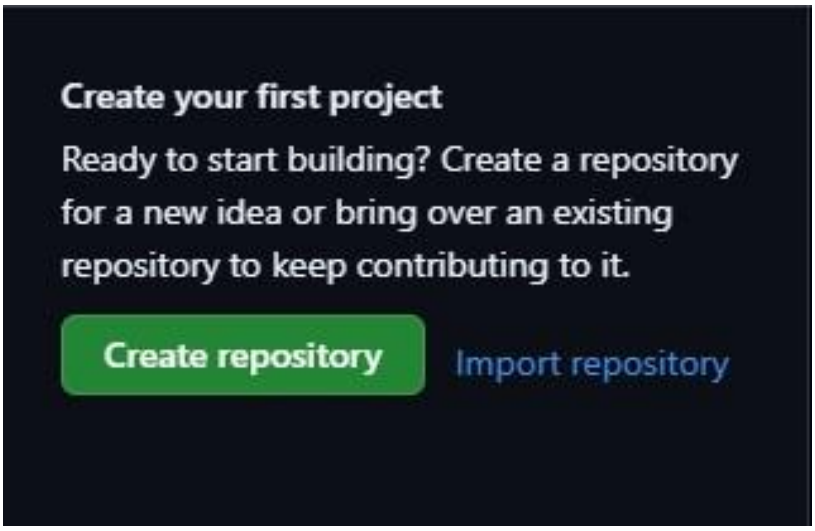
- Search for “GitHub” in any browser, then click on the first link (GitHub).



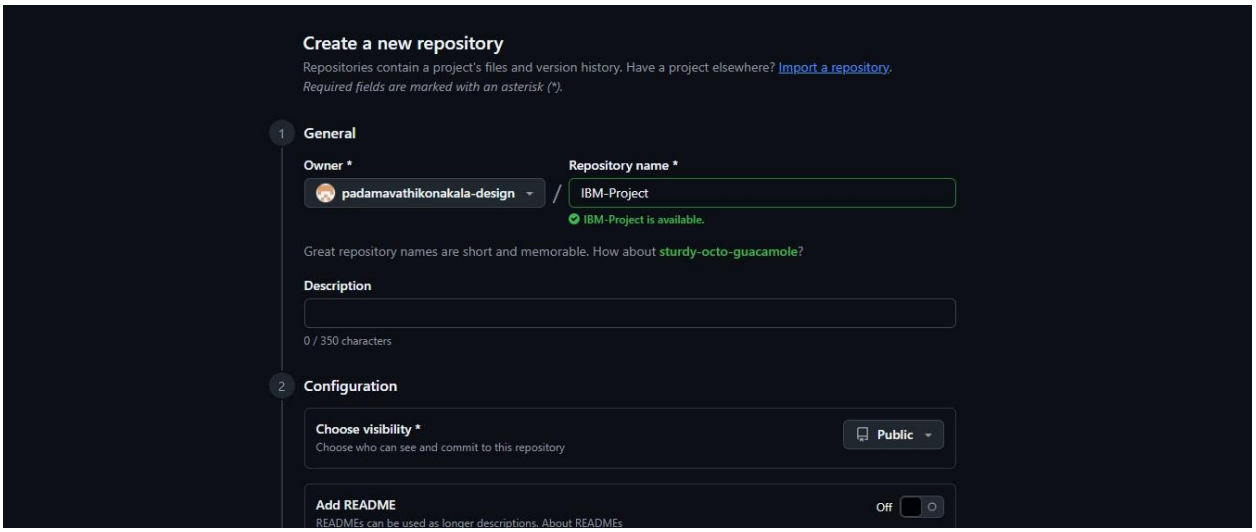
- Then click on “Signup” and create your own account in GitHub. If you already have an account click on “Sign in”



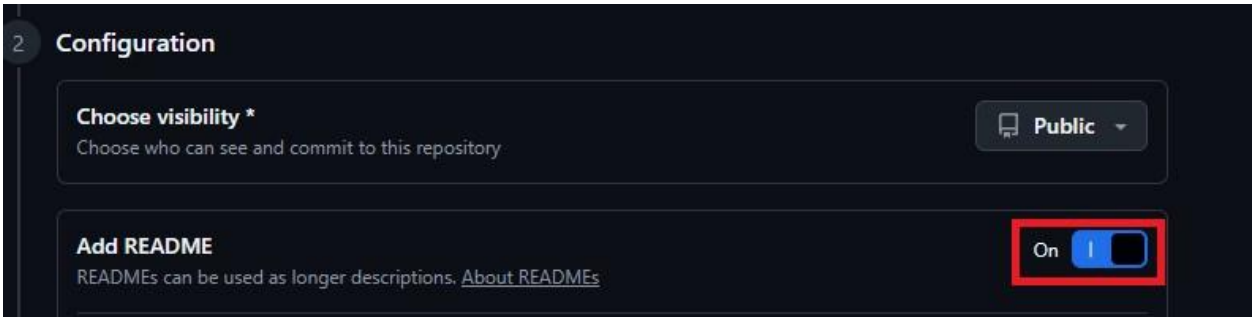
- Click on “Create repository”.



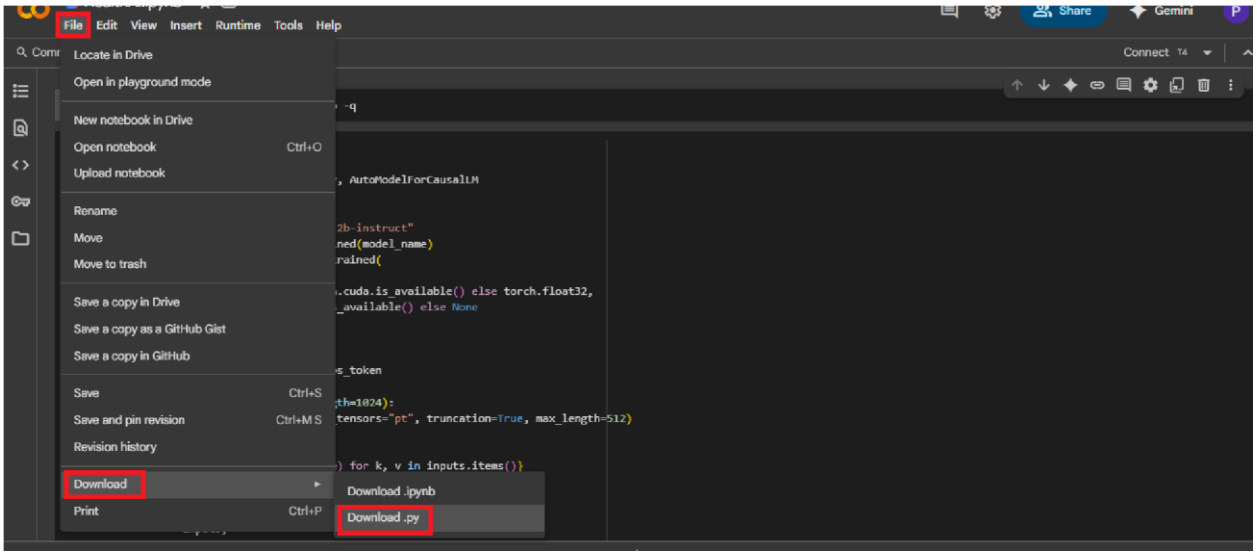
- In “General” Name your repo. (Here I have given “IBM-Project” as my repo name and it is available)



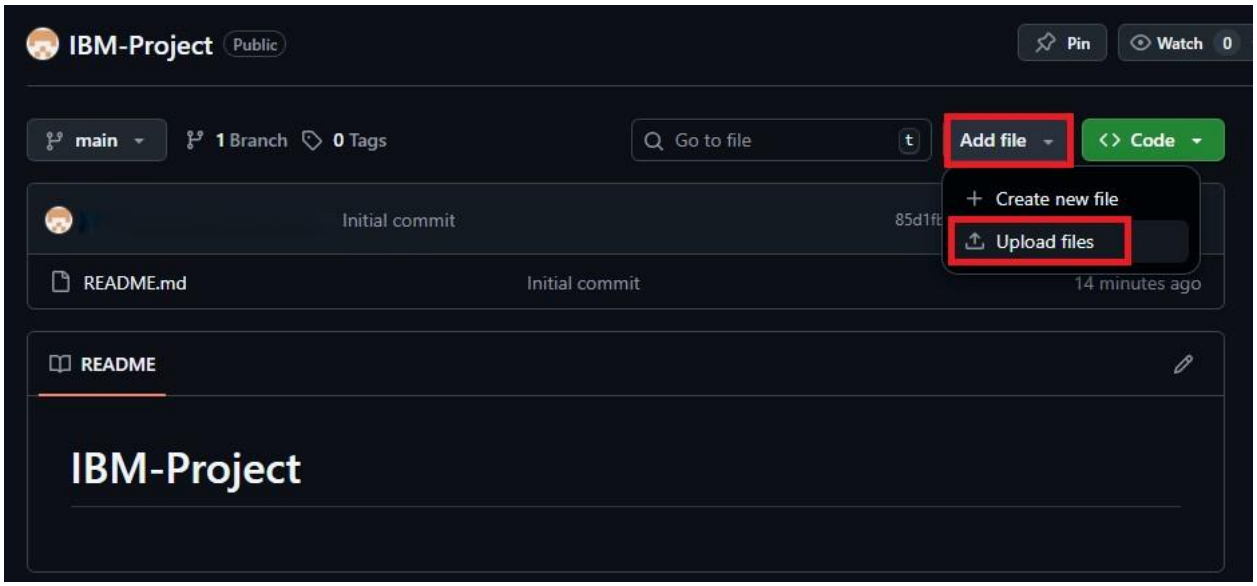
- In “Configurations” Turn On “Add readme” file Option.



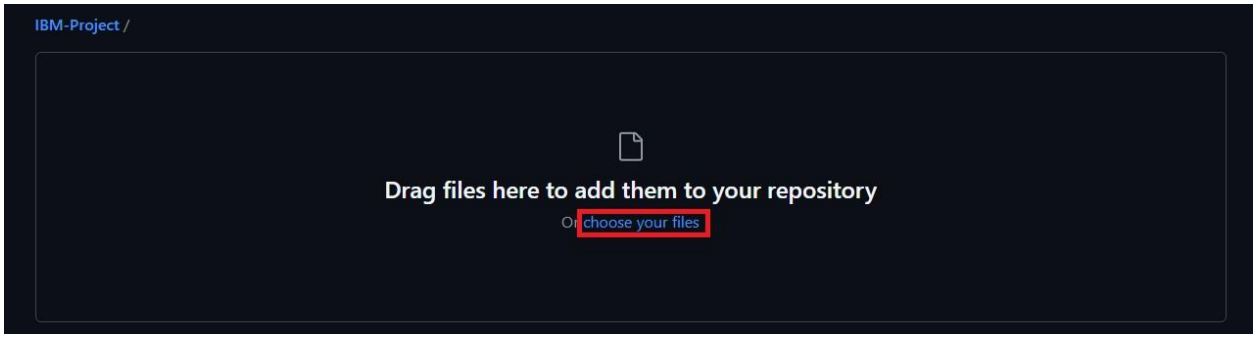
- Now Download your code from Google collab by Clicking on “File”, then Goto “Download” then download as “.py”.



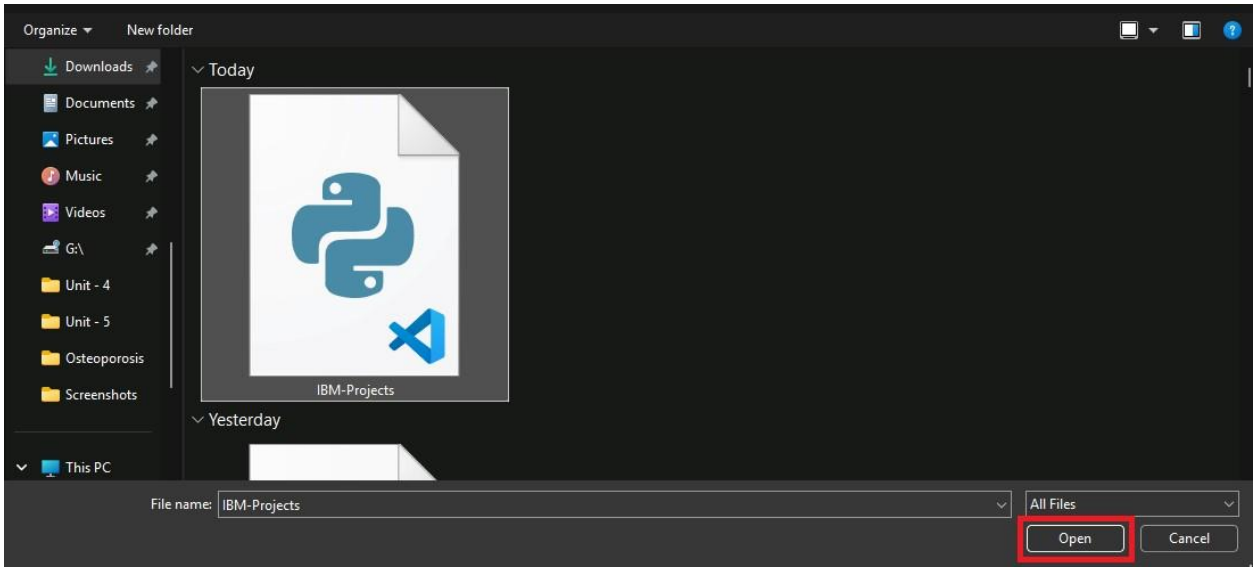
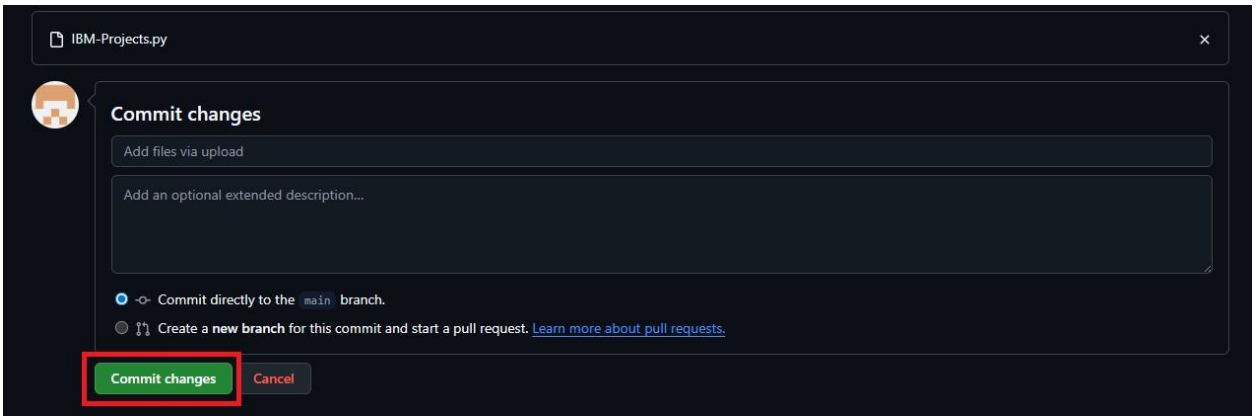
- Then your repository is created, then Click on “Add file” Option. Then Click “Upload files” to upload your files.



- Click on “choose your files”.



- Choose your project file and click on “Open”.



- After your file has Uploaded Click on “Commit changes”.